

SPRINGBANK, AB, Canada

WMO: 718600

Lat: 51.100N

Lon: 114.370W

Elev: 3940

StdP: 12.72

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-16.4	-31.0	1.1	-22.5	-24.1	1.7	-16.5	30.2	40.5	27.3	38.8	5.5	320	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	81.5	60.0	78.1	58.8	74.9	57.8	62.7	76.4	61.0	74.2	59.2	72.3	9.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.5	81.8	68.6	55.6	76.2	66.2	53.8	71.2	63.9	30.3	76.1	28.9	74.5	27.6	72.6	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.0	21.4	18.6	DB	-30.4	86.8	8.5	3.9	-36.6	89.6	-41.5	91.9	-46.3	94.1	-52.5	96.9	
			WB	-30.7	66.0	8.4	1.7	-36.7	67.3	-41.6	68.3	-46.3	69.3	-52.4	70.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	16.2	18.0	25.8	37.2	46.5	53.5	59.0	58.0	49.3	38.8	25.4	16.9
	DBStd	18.94	15.54	15.10	14.24	8.86	8.03	5.63	5.02	5.38	7.79	8.08	13.05	14.65
	HDD50	5500	1047	896	750	388	162	28	4	5	103	355	737	1025
	HDD65	10170	1512	1316	1215	833	573	346	194	221	471	812	1187	1490
	CDD50	817	0	0	0	4	54	133	282	254	81	8	1	0
	CDD65	13	0	0	0	0	0	2	7	4	0	0	0	0
	CDH74	872	0	0	0	3	40	72	323	362	72	0	0	0
	CDH80	159	0	0	0	0	1	11	57	79	11	0	0	0
Wind	WSAvg	7.8	7.6	7.7	7.8	8.6	8.7	8.3	7.3	6.9	7.2	8.2	7.3	8.0
Precipitation	PrecAvg	19.6	0.6	0.5	0.9	1.4	3.0	4.4	2.6	2.4	1.7	0.9	0.7	0.6
	PrecMax	27.0	1.1	1.0	2.3	3.3	5.3	13.8	6.3	4.9	4.3	2.1	2.2	1.9
	PrecMin	13.0	0.1	0.0	0.3	0.5	1.0	1.7	0.8	0.5	0.3	0.2	0.0	0.1
	PrecStd	3.3	0.3	0.3	0.5	0.7	1.3	2.4	1.4	1.2	1.0	0.5	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	52.2	60.9	70.8	78.5	81.1	85.5	86.4	80.5	69.6	58.6	50.9
		MCWB	38.2	39.5	43.2	49.4	55.0	60.7	61.9	60.9	55.8	49.7	44.2	37.4
	2%	DB	44.5	46.3	55.0	64.2	74.0	75.5	81.0	82.0	75.5	63.4	52.0	44.2
		MCWB	35.7	36.4	41.2	45.9	53.1	57.3	61.6	59.5	55.8	48.0	41.0	34.5
	5%	DB	41.5	42.0	49.8	59.0	69.2	71.5	77.7	78.1	71.0	58.3	47.1	40.3
		MCWB	34.0	33.8	38.6	43.6	51.0	55.6	60.3	59.1	54.2	44.9	37.8	32.5
	10%	DB	37.6	38.1	44.6	53.7	64.5	68.3	74.1	74.1	66.3	53.9	42.6	36.3
		MCWB	31.1	31.1	35.4	40.5	49.1	54.0	59.2	58.6	52.3	42.5	35.3	29.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	40.3	44.4	49.9	56.0	61.8	65.7	64.6	59.4	51.7	44.9	38.2
		MCDB	47.3	50.4	57.4	70.0	74.0	76.7	77.9	78.8	76.2	66.2	55.7	47.7
	2%	WB	36.4	36.7	41.9	46.9	54.1	59.0	63.4	62.2	56.5	48.5	41.6	35.3
		MCDB	43.9	45.5	54.3	62.5	70.7	71.6	76.6	75.9	72.2	61.4	51.0	43.5
	5%	WB	34.1	34.3	38.7	44.1	52.1	57.0	61.7	60.5	54.7	45.7	38.4	32.6
		MCDB	41.4	41.6	48.9	58.1	66.9	69.3	74.9	73.9	69.0	57.3	46.3	39.5
	10%	WB	31.2	31.4	35.6	41.2	49.9	54.9	60.0	58.9	52.7	43.0	35.2	29.9
		MCDB	37.4	37.4	44.4	52.3	62.9	66.1	72.6	72.7	65.1	53.3	42.2	36.0
Mean Daily Temperature Range	5% DB	MDBR	20.4	21.4	22.2	23.3	24.3	23.2	25.7	26.9	25.2	23.8	20.3	19.7
		MCDBR	23.8	24.4	28.8	33.2	34.4	31.1	33.3	36.2	36.2	30.8	26.5	22.7
		MCWBR	17.4	16.4	17.4	18.4	16.8	15.7	16.5	17.4	18.8	17.9	16.6	15.4
	5% WB	MCDBR	23.1	24.1	28.2	32.0	32.0	27.8	29.5	31.2	33.0	30.1	26.0	22.6
		MCWBR	17.3	16.8	17.5	18.2	16.1	14.8	15.7	16.2	17.9	17.8	16.7	15.9
Clear-Sky Solar Irradiance	taub	0.249	0.267	0.279	0.334	0.340	0.346	0.340	0.355	0.310	0.277	0.252	0.236	
	taud	2.443	2.446	2.478	2.326	2.376	2.395	2.416	2.387	2.485	2.562	2.510	2.426	
	Ebn at Noon	257	280	297	288	290	288	288	278	281	275	255	244	
	Edh at Noon	18	23	28	36	36	36	35	34	27	21	16	15	
All-Sky Solar Radiation	RadAvg	356	660	1048	1452	1699	1799	2007	1659	1169	725	385	279	
	RadStd	23	47	72	102	139	143	138	117	132	72	26	18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page